

Torsion groups and the Bienvenu–Geroldinger conjecture

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Abstract. Let M be a monoid (written multiplicatively). Equipped with the operation of setwise multiplication induced by M on its power set, the collection of all finite subsets of M containing the identity element forms a monoid itself, denoted by $\mathcal{P}_{\text{fin},1}(M)$ and called the reduced finitary power monoid of M .

This naturally leads to the question of whether, for all H and K in a given class of monoids, $\mathcal{P}_{\text{fin},1}(H)$ and $\mathcal{P}_{\text{fin},1}(K)$ are isomorphic if and only if H and K are. The problem originates from a conjecture of Bienvenu and Geroldinger [1], which was recently settled in [4]. In this talk, I will discuss how the problem has a positive solution in the case where H and K are both torsion groups.

About the speaker. Weihao Yan is a first-year PhD student in the School of Mathematical Sciences at Hebei Normal University, China. He is enrolled in the university’s combined Master’s–PhD program. During the Master’s phase of his studies, he conducted research on power semigroups under the supervision of Salvatore Tringali. He is currently working on geometric constructions of divisible design graphs and their applications to signed graphs under the supervision of Bart De Bruyn and Sergey Goryainov. He has published two papers on power semigroups in the Proceedings of the American Mathematical Society and the Journal of Combinatorial Theory, Series A, and has a third paper currently under review.

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